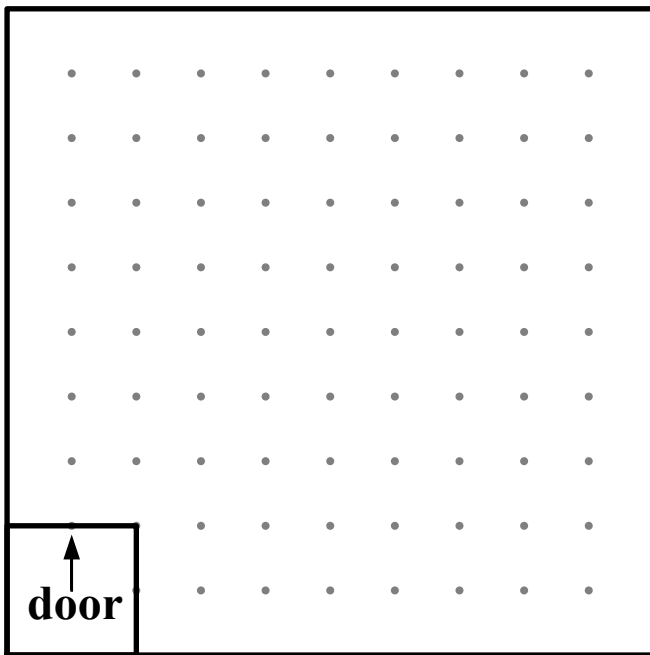


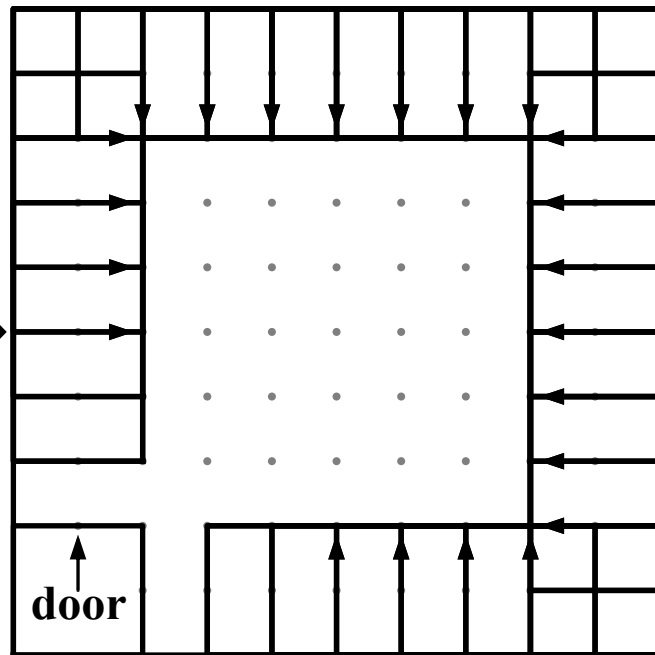
Initial State

Create grids in the room to reduce the search space.

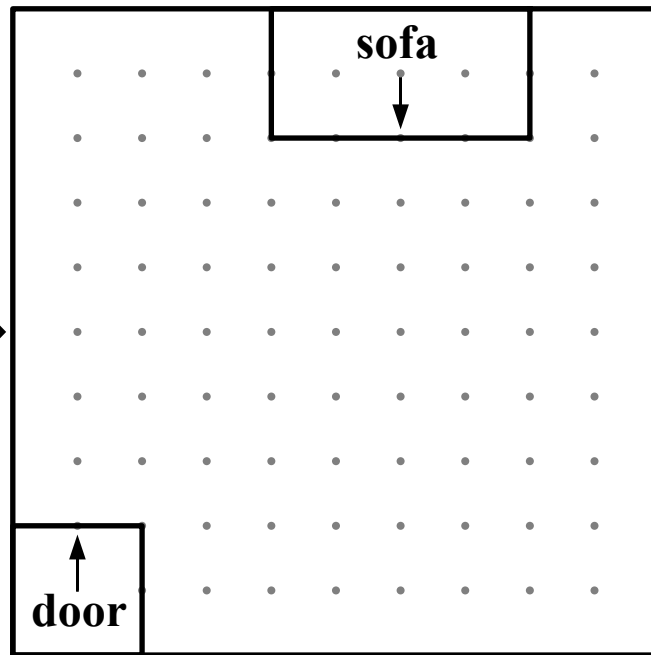


Place the anchor object sofa, at the edge of the room

Get all valid placements by satisfying the edge constraint.

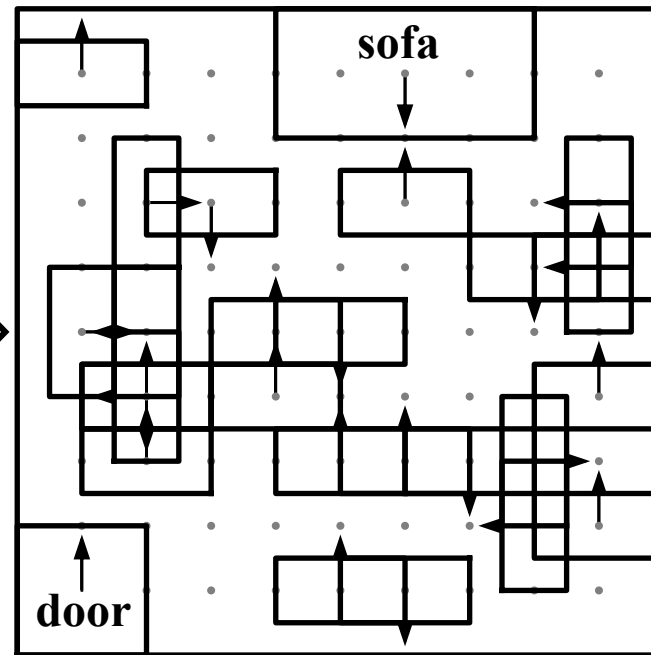


Randomly choose one placement for the anchor object.



Place the coffee table, in front of, face to, and near the sofa

Get all valid placements by satisfying hard constraints.



Choose the optimal placement which satisfies most constraints.

